A Scalable Method of Adaptive LVRT Settings Adjustment for Voltage Security Enhancement in Power Systems With High Renewable Penetration

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***Abstract*—Recent expansion of inverter-based renewable generation (RG) has introduced unprecedented stability issues. One major challenge stems from tripping RGs offline during disturbances based on predefined operation thresholds (ride through curves) on voltage magnitudes and frequency at the point of common coupling (PCC). The current practice uses a single set of thresholds for the whole year across all RGs in the system. This could result in loss of these resources during critical times, particularly endangering system voltage security, i.e a significant reduction in the load margin (LM). On the other hand, relaxing these limits could result in them staying online even during islanding scenarios, which is a safety concern to the restoration crew. Thus, an approach to find minimal relaxations to these limits to enhance voltage security is proposed here. This process first constructs an optimization model aiming to minimize adjustments on RG voltage limits to reach anticipated LM increase and then solves it with a multi-stage screening method including base power-voltage (P-V) curve acquisition, critical RGs identification, successive screenings of candidate solutions based on sensitivities, and optimal solutions validation. This largely increases the scalability of the algorithm, which is demonstrated on the IEEE 14-bus, 118-bus systems, and a 500-bus system.**

***Index Terms*—Adaptive control, voltage security, low-voltage ride-through, renewable energy, continuation power flow.**

I. Introduction

GROWING concerns on carbon footprint have urged electric power utilities to replace fossil fuel-based generation with renewable energy resources. The increasing integration of RGs [1], on the other hand, has made the system operation much more complicated. In the past, RGs were conservatively tripped offline during disturbances to avoid energizing islands. However, this practice posed a threat to system stability with the growing penetration of RGs. Thus, the time-dependent operation limits in the form of ride-through curves were introduced to allow them to remain online as long as there wasn't an unintentional islanding situation, in which case these limits were expected to be violated. These limits include the Low-Voltage Ride-Through (LVRT) curve and the Low-Frequency Ride-Through (LFRT) curve concerning the voltage and frequency at the PCC of each RG, respectively [2].

The LVRT settings are designed to ensure that RGs only trip offline during islanding scenarios [3]. However, in reality, it is extremely difficult to distinguish islanding cases based only on the local measurements at the PCC [4]. Furthermore, there is no widely accepted approach to arriving at an optimal setting. A conservatively designed LVRT setting can result in a widespread tripping event, especially for transmission network faults, which can threaten system stability, as happened in the United Kingdom on August 9th, 2019 [5]. Researchers in [6], [7] have talked about the negative effect of this generation loss on transient stability. Given low voltage scenarios are of major concern in both steady state and dynamical stability analysis, this article focused on the LVRT and considered its influence on system voltage security.

A widely used metric for voltage security is LM, which refers to the maximum load the system can supply with all the online resources. The CPFLOW [8] is the most reliable method to estimate it, however, is computationally intensive. Given a fixed load change direction, CPFLOW traces the one-dimensional P-V curve in the state space connecting the load flow solutions corresponding to increasing system loading. The curve is traced till the solution no longer exists, a condition referred to as the saddle-node bifurcation (SNB). There are also fast methods, such as [9], where a look-ahead approach is utilized to approximate the projection of this curve at one or more critical buses by using a few load flow solutions. Lately, holomorphic embedding techniques[10] have become more popular which provides a closed-form expression for this curve of load flow solutions as a function of loading. Note that the load flow equations can undergo structural changes on generators hitting reactive power limits which the existing methods are capable of dealing with. However, the case of RGs tripping offline is slightly different in that it introduces a discontinuity (jump) in the curve itself, which has not been studied in the previous work. Data-driven methods [11]–[13] are also proposed in recent years, which provide an estimate of the LM given measured system states. But they aim more on the online applications and heavily depend

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on measurements. The results are either probabilistic or a range where the LM locates, which is not applicable here.

Given a scenario with rich RG penetration and low load margin, the next step is to understand mitigation approaches to enhance it. Existing methods to enhance the LM mostly focus on generation re-dispatch[14], load control [15], shunt capacitance placement [16], network topology optimization[9], and other conventional perspectives. However, situations can arise when they are not sufficient especially when there is a threat of losing multiple critical RGs[17] that will significantly reduce the load margin. Under those circumstances, there may be a need to temporarily relax the LVRT constraints for maintaining system voltage security. It is important to clarify that the authors acknowledge the criticality of LVRT's function to avoid situations with RGs staying online in an islanding scenario. Therefore, the goal of this study is to achieve certain LM improvement through minimal adjustment in LVRT settings and implement it only when necessary. Even if not used for online applications, this tool can help identify areas requiring additional reactive power support to reduce the amount of generation lost.

Existing research on LVRT setting adjustment mainly concentrates on designing control schemes to enhance the RGs' LVRT capability [18]–[20]. Others propose control strategies to provide better reactive support [21], [22] or to enhance system small signal stability[23]. Researchers in [24] focus on equipping the inverter itself with autonomous model predictive control and enable the inverter to switch between normal operation and fault modes. The fault ride-through capability of RGs is enhanced by autonomous P/Q setpoint changes. Additive devices like energy storage [25], [26] and various fault current limiters [27], [28] are also integrated to limit the fault current and further enhance the fault ride-through capability of RGs. For the LVRT criterion, researchers in [29] attempted to summarize proper LVRT settings from real-world practices. [30] proposed a method to justify the fault ride-through requirements for newly integrated RGs in the system based on identified critical contingencies analysis. Although this work did not provide quantitative methods to find proper ride though requirements, it confirms the necessity to find more adaptive RG LVRT settings for better voltage security and more efficient renewable energy utilization. However, all these works treat the LVRT settings as an existing standard; they do not attempt to adjust the criterion for more secure and efficient system operation.

In this paper, we proposed a scalable methodology to adaptively adjust RGs' LVRT settings to improve the voltage security of a system with high renewables penetration. This multi-stage approach involves identifying critical RGs in the system given certain operation condition, filtering feasible schemes of RGs blocked from tripping to reach desired LM, selecting optimum feasible solutions by estimating LVRT setting adjustment quantities, and evaluating the selected solutions to find true LM increment and LVRT settings adjustment as results. With the measures to scale down the problem and multiple estimation steps, the proposed method is fast and suitable for large-scale systems. The main contributions include:

1. Proposed the novel problem of RGs flexible operation for voltage security of system with high renewable penetration to facilitate renewable energy transition.
2. Introduced a novel voltage prediction method for the identification of critical RGs to boost efficiency.
3. Constructed a scheme combining multiple estimation-based analytic tools to find effective and feasible solutions of RG LVRT limit adjustment for system LM improvement.
4. Applied the proposed method to large scale renewable-rich systems and validated its capability and scalability.

This paper is an extension of the work in [31] which proposed the concept about improving system voltage security through flexible operation of RGs. To extend the proposed concept and implement it with scalability and robustness, on the basis set by [31], the authors introduced several improvements to the intermedia steps and, more importantly, following advancement in the research: 1) further detailed and formalized the research problem and explained key concepts like the *Base Curve* and the discrete system condition change brought by renewables tripping; 2) improved the proposed voltage prediction method by introducing Least Squares Estimation (LSE) for robustness and curvature relevant data point selection method for better accuracy; 3) parallelized the sensitivity based LM estimation method to improve the method scalability; 4) proposed continuous RG connection statuses instead of binary variables in the LVRT constant estimation step to mitigate Newton-Raphson (NR) divergence issue; 5) deployed the proposed method in large scale RG-rich study cases and reached expected results.

The rest of the paper is organized as follows: Section II discusses the impacts of RG tripping by LVRT violation to system voltage security; Section III provides a theoretical formulation of the problem studied; Section IV details the proposed adaptive LVRT adjustment methodology; Section V presents the case study results and discussions on IEEE 14-bus system, IEEE 118-bus systems, and a 500-bus system; Section VI concludes the whole paper.

II. Impacts of RG LVRT Violations on Load Margin

Tripping of RGs leads to a sudden increase in the loading, thus reducing the load margin. Since this tripping is governed by the voltage value at the PCC, the LVRT curve plays a significant role in voltage security.

A. Voltage Security and Load Margin

LM can be calculated through CPFLOW. The standard power flow model is shown in (1),

$$f\ (\boldsymbol{x}, \lambda) = F(\boldsymbol{x}) + b\lambda\ = 0 \tag{1}$$

where $\boldsymbol{F}(\boldsymbol{x})$ is the vector of the difference between power supply and demand; $\lambda \in \mathbb{R}$ is the total load scaling factor; $\boldsymbol{b} \in \mathbb{R}^{N_L}$ is the direction of load growth; N_L is the number of loads; and $\boldsymbol{x} = [\mathbf{V}\ \boldsymbol{\theta}]^T$ is the vector of network states comprising of the voltage magnitudes $\mathbf{V}$ at all PQ buses and phase angles $\boldsymbol{\theta}$ at all PV and PQ buses.

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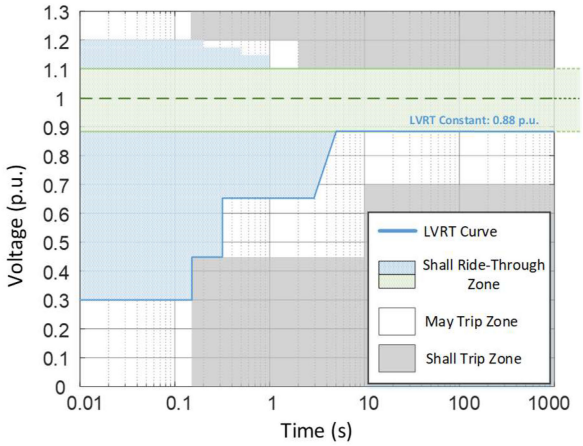


Fig. 1. IEEE Std. 1547 Category 2 Voltage Ride-Through curve.

The equilibrium points of the system are the states that satisfy (1) for a given λ. On continuously varying λ, the equilibrium point(s) trace one-dimensional solution curves in the state-space. Of particular interest is the curve passing through the current operating point ($\lambda = \lambda_0$). The maximum value of λ (load) for which a solution to (1) exists is the LM, which is a widely used metric for voltage security [9]. It describes the distance from the current operating point to voltage collapse and thus can quantify the capability of the system to maintain voltage stability. Beyond the LM, no solution exists for (1). The operation condition at this SNB point where $\lambda = \mathrm{LM}$ is characterized by the singularity of the Jacobian matrix $\partial \boldsymbol{f}/\partial \boldsymbol{x}$. The graph of voltages vs. λ along this solution curve is called the P-V curve.

B. P-V Curve Under LVRT Tripping

A typical LVRT curve represents a time-dependent lower limit on voltage magnitude at the PCC of an RG. The violation of this limit results in tripping the RG offline. Note that the standards governing these limits do provide flexibility in their settings[2], as shown in Fig. 1 to allow for differences in voltage recovery among different networks or parts of the same network. The 'Shall Trip Zone' in Fig. 1 indicates the RG should be strictly tripped offline when the voltage reaches it at the corresponding time. The 'Shall Ride-Through Zone' is the recommended (while conservative) tripping limit for safe operation. The region in between is labeled as 'May Trip Zone' where there is flexibility for LVRT adjustment. However, from the safety perspective, it is important to ensure that the chosen settings show as violated during islanding scenarios. Utilities tend to use certain default setting throughout their networks, at times without proper justification. The present work proposes an approach to arrive at minimal adjustments to ensure both safety and system reliability. It is important to clarify here that since we are looking at the steady-state voltage stability, this article will be focusing on adjusting the final constant part value of the curve. The steady state voltage at PCC being lower than this constant will trigger the RG to trip.

Since the connection status of RGs can be represented by discrete variables, the overall system is modeled as a hybrid system,

$$f\ (\boldsymbol{x}, \lambda, \boldsymbol{\Gamma}_{RG}) = F\,(\boldsymbol{x}, \boldsymbol{\Gamma}_{RG}) + b\lambda \tag{2}$$

where $\boldsymbol{\Gamma}_{RG} \in \mathbb{Z}_2^{N_{RG}\times 1}$ represents the connection statuses (0–offline/1–online) of RGs; N_{RG} is the number of RGs. Clearly, there are $2^{N_{RG}}$ combinations of possible RG connection statuses with their associated P-V curves and the system will jump/switch from one curve to the other on violation of LVRT limit(s). For example, Fig. 2(a) shows all four possible P-V curves of a system with two RGs. Each curve is labeled with a corresponding $\boldsymbol{\Gamma}_{RG}$. For instance, the blue dashed curve '01' illustrates the P-V curve with RG 1 always offline and RG 2 always online. It worth noting that with the loss of more RGs, the system generally has smaller LMs (the SNB of the corresponding curve is closer to zero) because of the equivalent increase in system demand. As shown in Fig. 2(a), the '*Base Curve*' (blue bold curve) depicts the voltage variation at PCC of RG1 given RGs' original LVRT settings. RG 1's LVRT limit is set to be 0.95 p.u. (red dash-dot line). In the beginning, both RGs are online and system states stay on the curve '11' (black dashed curve). Once the voltage reaches 0.95 p.u. where λ is around 0.4, RG1 is tripped offline and the system states jump from the curve '11' to curve '01' (green dashed curve). Fig. 2(b) illustrates this event in more detail. In order to estimate the load margin for such a system, a slight modification to the conventional CPFLOW is proposed.

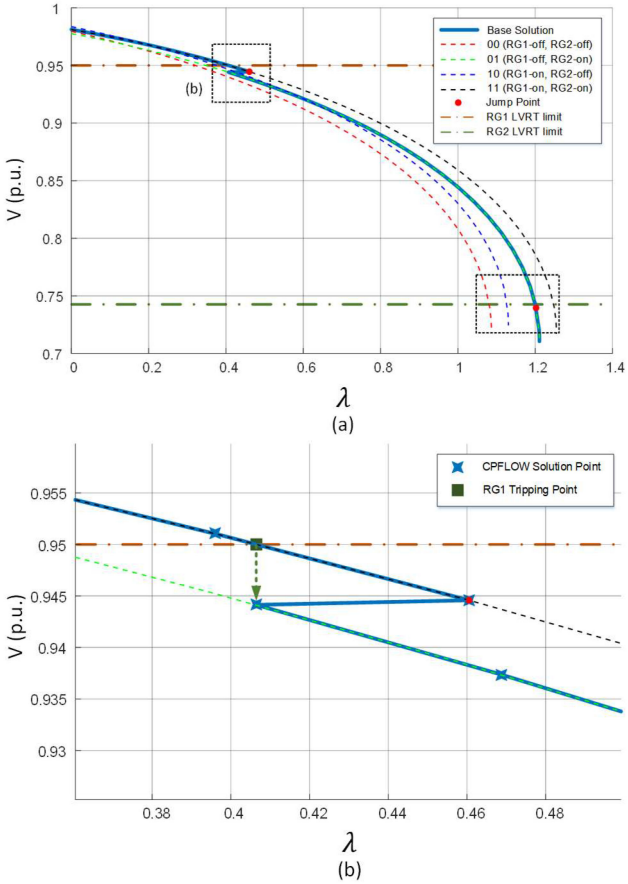


Fig. 2. Renewables tripping by violation of LVRT limit.

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Since there is no way to anticipate the loading λ at which an LVRT limit will be violated, the voltage magnitude V on the RG bus is continuously observed. If V is found below RG's LVRT limit, the CPFLOW stops and the current states are collected. Using current states, the state values at the previous CPFLOW step, and the LVRT limit value, one can estimate the λ value corresponding to the violation of LVRT through interpolation. Corresponding states on the new curve can be further found through power flow equations. This violating λ at the jump is determined by both local voltage support and LVRT settings.

Another interesting scenario that can occur is the loss of equilibrium immediately following such a jump. This happens when the succeeding PV curve has a load margin lower than the violating λ. As shown in Fig. 2(a) at the bottom right where the LVRT limit on RG2 is violated and the system is ought to jump from curve '01' to '00' (red dashed). However, the later curve has already reached its collapsing point (SNB). The tripping of RG2 will cause instantaneous voltage collapse. Therefore, the only way to find the LM or let the P-V curve reach the SNB is to block the corresponding RG from tripping.

It is to be noted that there exists another set of discrete variables corresponding to generators hitting their reactive power output upper or lower limits. This kind of scenarios are considered in the analysis by altering the bus type and resetting its PQ loads but omitted from presentation here.

III. Problem Formulation

The goal of this research is to achieve an anticipated LM $\lambda = \lambda_{\text{limit}}$ through minimum adjustment of LVRT limits. This translates to having a load flow solution at $\lambda = \lambda_{limit}$ by stopping a few RGs from tripping offline to maintain enough generation. To achieve this, the new LVRT settings at those RGs can be set to be lower than their corresponding voltage values at $\lambda = \lambda_{\text{limit}}$. Therefore, the proposed adaptive RG LVRT adjustment problem can be formulated as an optimization problem,

$$\underset{\Delta \boldsymbol{c}_{LVRT}}{\operatorname{argmin}} \sum \Delta \boldsymbol{c}_{LVRT}$$

$$s.t. \qquad \Delta \boldsymbol{c}_{LVRT}^{(i)} \geq 0, \ \ \forall i \in \{1, 2, \ldots, N_{RG}\} \quad (a)$$

$$\exists \boldsymbol{x} \quad s.t. \ \ \boldsymbol{f}\left(\boldsymbol{x}, \ \lambda_{\text{limit}}, \ \boldsymbol{\Gamma}_{RG}\left(\boldsymbol{x}, \ \boldsymbol{c}_{LVRT}^{original} - \Delta \boldsymbol{c}_{LVRT}\right)\right) = 0 \quad (b) \tag{3}$$

where $\Delta \boldsymbol{c}_{LVRT} = \boldsymbol{c}_{LVRT}^{original} - \boldsymbol{c}_{LVRT} \in \mathbb{R}^{N_{RG} \times 1}$ is the vector of LVRT adjustments for all the RGs in the system; $\boldsymbol{c}_{LVRT}^{original} \in \mathbb{R}^{N_{RG} \times 1}$ is the vector of original LVRT settings before any adjustment; $\boldsymbol{c}_{LVRT} \in \mathbb{R}^{N_{RG} \times 1}$ is the vector of the aimed LVRT settings which is also the $\mathbf{V}_{RG}$ at $\lambda = \lambda_{\text{limit}}$; $\lambda_{\text{limit}} \in \mathbb{R}$ is the desired lower limit on LM. Since the connection status $\boldsymbol{\Gamma}_{RG}$ depends on the voltage magnitude and the LVRT setting, it can be written as:

$$\boldsymbol{\Gamma}_{RG}^{(i)} = \begin{cases} 0, V_j < \boldsymbol{c}_{LVRT}^{(i)} \\ 1, \quad \text{otherwise} \end{cases} \tag{4}$$

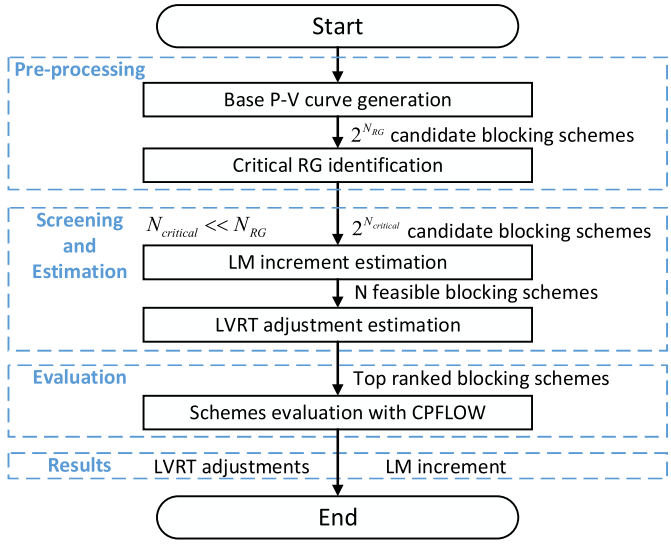


Fig. 3. Flow chart of the proposed method.

where the ith RG is installed on the jth bus; V_j is the voltage magnitude of bus j; and $c_{LVRT}^{(i)}$ is the LVRT setting of RG i.

Constraint (a) in (3) ensures that the adjustments are only reductions on original LVRT settings. Constraint (b) specifies that the LM of the system after the adjustments satisfies the aforementioned equilibrium/solution existence at $\lambda = \lambda_{\text{limit}}$.

IV. Methodology for Adaptive LVRT Adjustment

Given the nature of the variable $\boldsymbol{\Gamma}_{RG}$, the problem defined in (3) is a nonlinear mix-integer programming problem. It is not preferable and realistic to directly solve this problem, especially in a large-scale system. This is because checking the feasibility of the candidate solution, $\Delta \boldsymbol{c}_{LVRT}$, on constraint (b) involves an extremely heavy computational burden brought by CPFLOW-like processing. This is because each candidate solution decides the $\boldsymbol{\Gamma}_{RG}$ at λ_{limit} and $\boldsymbol{\Gamma}_{RG}$ further decides on which P-V curve the system ended up with. As shown in Fig. 2, different P-V curves most likely associate with various LMs. For different LVRT settings, there is no way other than full CPFLOW to find the actual tripping events the system is going to experience and thus corresponding $\boldsymbol{\Gamma}_{RG}$. Therefore, using $\boldsymbol{\Gamma}_{RG}$ as a pivot, the problem can be solved in a two-step approach where the first step is to identify RGs that need to stay online so that the load margin is beyond λ_{limit}. This is followed by computing LVRT settings to ensure that scenario.

To solve this effectively and efficiently, this article propose a method that successively screens out infeasible/ineffective candidate solutions to focus on and evaluate only a handful of good solutions in greater detail. The method is composed of five steps (as shown in Fig. 3): 1) Run a full CPFLOW with RGs tripping based on original LVRT settings to get the base P-V curve; 2) Find critical RGs in the system whose LVRT settings must be considered for adjustment; 3) Find effective combinations of RGs to be kept online/blocked from tripping offline to meet the LM limit; 4) Find LVRT adjustments necessary to achieve

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these blockings to reach the least adjustment feasible solutions; 5) Evaluate the effectiveness of top-ranked solutions and their corresponding new LVRT settings using full CPFLOW. Each step is discussed in detail in the following subsections.

A. Identifying Critical RGs

A renewable-rich system generally contains a large quantity of RG buses and thus enormous combinations of RGs binary connection statuses. This could lead to an extremely large solution search space. For example, if there are 10 RGs installed in the system, the search space is composed of 2^{10} combinations of RGs to be blocked from tripping by adjusting their LVRT limits, namely blocking schemes. So, there is a need to find the critical RGs that tend to trip offline for a given load increase direction. This is to improve the efficiency by filtering out RGs that are unlikely to be tripped offline and thus non-sensitive to LM increment. If only 5 of these 10 RGs are critical (or tend to trip) the search space is dramatically decreased to 2^5.

The method to identify critical RGs is composed of two parts. Firstly, RGs whose LVRT settings are violated in the base P-V curve are critical by default. Besides those, other RGs whose V at $\lambda = \lambda_{\text{limit}}$ are lower than respective LVRT settings are also deemed critical. These V cannot be found directly given these RGs should have already been tripped offline before reaching λ_{limit}. Therefore, we assume all LVRT settings are temporarily disabled, i.e., all RGs are blocked from tripping at this stage.

Now, in order to avoid repeatedly running a CPFLOW, we used LSE to estimate the voltages at different RGs when $\lambda = \lambda_{limit}$. This is done by assuming a quadratic relationship between voltage at each RG bus and system loading λ as follows,

$$\left[\lambda^2 \ \lambda V_i \ V_i^2 \ \lambda \ V_i \ 1\right] \left[\alpha_i \ \beta_i \ \gamma_i \ \sigma_i \ \eta_i \ \mu_i\right]^{\mathrm{T}} = 0 \qquad (5)$$

where V_i is the voltage of bus i; $[\alpha_i \ \beta_i \ \gamma_i \ \sigma_i \ \eta_i \ \mu_i]^{\mathrm{T}}$ is the parameter vector of the corresponding quadratic. This assumption is in line with the existing methods to predict P-V curve [32], [33]. It has been seen that simpler models [32] only perform well for approximating the PV curve away from the SNB, while [33] only performs well for approximating closer to the SNB. Using a full quadratic relationship such as the one proposed tends to work well in both the scenarios and therefore is proposed here.

The estimation is based on the points, or (λ, V_i) pairs, on the first part of the *Base Curve* before the first RG tripping event. The LSE uses at least 6 points to form the data matrix for valid estimation results. The reverse Fibonacci sequence is recommended to introduce more diversity without losing the curvature information. To select these points on the *Base Curve*, take the point where the first trip event happens as the reference point, trace back towards the initial point of the *Base Curve* to find the closest point to this reference. This serves as the 1st point. Similarly, the 2nd, 3rd, 5th, and 8th points are found. Combining these five points with the initial point of the *Base Curve* to form the data matrix. This way the former mentioned three point maintain more curvature information because they are closer to the SNB and the latter three points are comparatively far away from the trip point which introduces diversity to the data to avoid LSE deficiency. Algorithm 1 shows the detailed method.

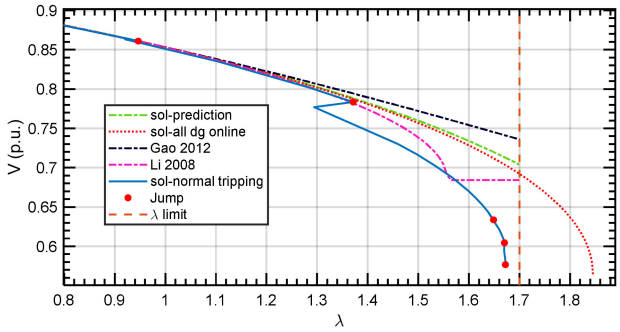


Fig. 4. Example RG bus full P-V curve prediction results and comparison.

Algorithm 1: Critical RGs Identification.

Input: RG buses set $\mathbb{M}$; RG bus Base Curves set, $\mathbb{K}$; RG bus LVRT limits set, $\mathbb{C}$; $\boldsymbol{\lambda_{limit}}$

Output: Critical RG buses set, $\mathbb{M}_{\boldsymbol{critical}}$

Initialization:

$\mathbb{M}_{\boldsymbol{critical}} = \{$RG buses whose LVRT limits were violated in Base Curves$\}$ Loop through remainder RG buses:

for $\boldsymbol{i^{th}}$ bus $\in (\mathbb{M} \setminus \mathbb{M}_{\boldsymbol{critical}})$ **do**
  find corresponding bus number, $\mathbb{M}(\boldsymbol{i}) = \boldsymbol{j}$
  find corresponding Base Curve, $\mathbb{K}(\boldsymbol{i}) = \mathcal{K}_{\boldsymbol{i}}$
  find corresponding LVRT limit, $\mathbb{C}(\boldsymbol{i}) = \boldsymbol{c}_{\boldsymbol{LVRT}}^{(\boldsymbol{i})}$
  locate $\boldsymbol{n^{th}}$ point on $\mathcal{K}_{\boldsymbol{i}}$ indicating first LVRT violation
  from data matrix with $1^{\boldsymbol{st}}$, $(\boldsymbol{n}-1)^{\boldsymbol{th}}$, $(\boldsymbol{n}-2)^{\boldsymbol{th}}$, $(\boldsymbol{n}-3)^{\boldsymbol{th}}$, $(\boldsymbol{n}-5)^{\boldsymbol{th}}$, $(\boldsymbol{n}-8)^{\boldsymbol{th}}$ points on $\mathcal{K}_{\boldsymbol{i}}$ and apply LSE to estimate the parameter vector in (5), $\boldsymbol{\nu_i}$
  use $\boldsymbol{\nu_i}$ to calculate voltage at $\boldsymbol{\lambda_{limit}}$, $\boldsymbol{V}_{\boldsymbol{est.}}^{(\boldsymbol{j})}$
  if $\boldsymbol{V}_{\boldsymbol{est.}}^{(\boldsymbol{j})} < \boldsymbol{c}_{\boldsymbol{LVRT}}^{(\boldsymbol{i})}$ **then**
    append $\boldsymbol{j}$ to $\mathbb{M}_{\boldsymbol{critical}}$
  end if
end for
return $\mathbb{M}_{\boldsymbol{critical}}$

An example of predicting the PV curve with all RGs online using the proposed method is shown in Fig. 4. To demonstrate the effectiveness of the proposed method (green dashed), two other methods that can be found, Gao 2012 [32] (black dot-dashed) and Li 2008 [33] (pink dot-dashed) are also implemented and results compared. As shown in the plot, there is not enough curvature in the Gao 2012 method using an Auto-Regressive model and three previous V values. Too much curvature is added in the Li 2008 method which tries to fit the curve using a parabola. The curve predicted with the proposed method is the closest to the target red dotted curve with all RGs online.

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B. Feasible Blocking Schemes Screening With LM Sensitivity

As discussed before, we first estimate the set of RGs to be kept online to enable reaching the load margin limit. Naturally, there will be certain combinations of RGs which are ineffective in achieving that limit. At this stage, we estimate how much each combination of RGs can enhance the load margin, if kept online/blocked from tripping offline. Let, $\mathbf{\Gamma}_{RG}^{(i)} = 1$ in a blocking scheme denote that the ith RG is blocked from tripping and '0' means otherwise. Since we only focus on critical RGs (identified in the previous step), entries in $\mathbf{\Gamma}_{RG}$ corresponding to non-critical RGs are set to be '0'.

In this work, we model RGs as negative loads. To get a conservative estimate for the effectiveness of a blocking scheme, we model the condition of not changing the LVRT setting (or not blocking) as a loss of active power injection. This is to avoid running full CPFLOW for each blocking scheme. The power flow equations are linearized at the *Base Curve* SNB. Using the sensitivity-based method proposed in [34], one can find the sensitivity of LM w.r.t. change in connection status of each RG from the *Base Curve*. This sensitivity is obtained with the derivative of the system power flow function at the SNB,

$$\left.\frac{\partial f}{\partial \boldsymbol{x}}\right|_{*} dx + \left.\frac{\partial f}{\partial \lambda}\right|_{*} d\lambda + \left.\frac{\partial f}{\partial \mathbf{\Gamma}_{RG}}\right|_{*} d\,\mathbf{\Gamma}_{RG} = 0 \tag{6}$$

where $\bullet|_{*}$ indicates the derivative evaluated at the SNB. By pre-multiplying one left zero eigenvector $\upsilon|_{*}$ of $\frac{\partial f}{\partial x}|_{*}$, the first term on the left-hand side is eliminated and the desired sensitivity is found.

$$d\lambda = -\left(\upsilon \left.\frac{\partial f}{\partial \lambda}\right|_{*}\right)^{-1} \left(\upsilon \left.\frac{\partial f}{\partial \mathbf{\Gamma}_{RG}}\right|_{*}\right) d\,\mathbf{\Gamma}_{RG} = A\, d\mathbf{\Gamma}_{RG} \tag{7}$$

For the jth blocking scheme, the estimated LM $\lambda_{j,est}^{*}$ is,

$$\lambda_{j,est}^{*} = \lambda_{base}^{*} + A\,\Delta\mathbf{\Gamma}_{\mathrm{RG}} \tag{8}$$

where λ_{base}^{*} refers to the LM of the *Base Curve*; $\Delta\,\mathbf{\Gamma}_{RG} = \mathbf{\Gamma}_{RG,j} - \mathbf{\Gamma}_{RG,base}$ refers to the RG statuses' change from the final RG connecting statuses $\mathbf{\Gamma}_{RG,base}$ in the base case to the concerned blocking scheme $\mathbf{\Gamma}_{RG,j}$. Each blocking scheme whose $\lambda_{j,est}^{*} \geq \lambda_{\mathrm{limit}}$ is considered feasible. All the feasible blocking schemes are passed to the next step for quantitative LVRT adjustment estimation. The detailed algorithm is shown in Algorithm 2.

C. Estimating Adjustment to LVRT Limits

Previous screening identifies feasible RG blocking schemes. For large-scale renewable-rich systems, there could still be a large number of feasible schemes after this filtering which are time-consuming to evaluate one by one. Therefore, further screening is needed to find the most secure feasible schemes worth the computation effort of evaluation with CPFLOW. The security here means a minimum adjustment to LVRT limits. A method is proposed here to estimate the voltages of the RG PCCs at λ_{limit} given blocking schemes and further to find the reduction compared to the original LVRT limits. This iterative method starts from known equilibrium points on the *Base Curve* which is the only accurate information we have as of now. Candidate blocking schemes can be mapped to the curves w.r.t. the RG connecting statuses. System states at $\lambda = \lambda_{\mathrm{limit}}$ on these curves are the focuses.

Algorithm 2: Feasible Blocking Schemes Screening.

Input: Critical RG buses set, $\mathbb{M}_{\boldsymbol{critical}}$; Base Curve RG final statuses, $\mathbf{\Gamma}_{\boldsymbol{RG,base}}$; Base Curve LM, $\boldsymbol{\lambda}_{\boldsymbol{base}}^{*}$; $\boldsymbol{\lambda}_{\boldsymbol{limit}}$
Output: Feasible blocking schemes set, $\mathbb{B}_{\boldsymbol{feas}}$

Generate candidate blocking schemes set, $\mathbb{B}$:
find number of critical RGs from $\mathbb{M}_{\boldsymbol{critical}}$, $\boldsymbol{N_{RG,c}}$
generate all binary combinations of $\boldsymbol{N_{RG,c}}$ bits, save to set, $\hat{\mathbb{B}}$
for $\boldsymbol{i^{th}}$ combination $\in \hat{\mathbb{B}}$ **do**
  add '0's into $\hat{\mathbb{B}}(\boldsymbol{i})$ at positions corresponding to non-critical RGs to form candidate blocking scheme, $\mathbf{\Gamma}_{\boldsymbol{RG,i}} \in \mathbb{Z}_2^{N_{RG}\times 1}$
  append $\mathbf{\Gamma}_{\boldsymbol{RG,i}}$ to $\mathbb{B}$
end for
Loop through candidate blocking schemes:
for $\boldsymbol{j^{th}}$ scheme $\in \mathbb{B}$ **do**
  find difference on RG connection statuses between current scheme and Base Curve, $\Delta\,\mathbf{\Gamma}_{\boldsymbol{RG}} = \mathbf{\Gamma}_{\boldsymbol{RG,j}} - \mathbf{\Gamma}_{\boldsymbol{RG,base}}$
  estimate LM correspond to the current scheme, $\boldsymbol{\lambda}_{\boldsymbol{j,est}}^{*}$, using (8) and $\boldsymbol{\lambda}_{\boldsymbol{base}}^{*}$
  if $\boldsymbol{\lambda}_{\boldsymbol{j,est}}^{*} > \boldsymbol{\lambda}_{\boldsymbol{base}}^{*}$ **then**
    append $\mathbb{B}(\boldsymbol{j})$ to $\mathbb{B}_{\boldsymbol{feas}}$
  end if
end for
return $\mathbb{B}_{\boldsymbol{feas}}$

There are two scenarios to be considered depending on whether the base P-V curve has passed λ_{limit} or not, which determines the starting point of the process.

1) The Base Curve has Passed the LM Limit: For each feasible blocking scheme, the NR method is used based on the power flow equations with aimed RG connection statuses in the scheme, $\mathbf{\Gamma}_{RG,aimed}$. The starting point, $\boldsymbol{x}_{base}$, of the NR process is on the known base P-V curve, $\mathbf{\Gamma}_{RG,base}$, at $\lambda = \lambda_{\mathrm{limit}}$. The calculation evolves from $\boldsymbol{x}_{base}$ to the aimed states $\boldsymbol{x}_{aimed}$ with λ kept constant. The variables are the RG connection statuses. They are binary by nature, but to avoid NR process collapses, they can be taken as continuous variables in practice. This is equivalent to RG not operating on full output and thus is physically reasonable. The resulted state point $\boldsymbol{x}_{aimed}$ provides estimated voltages at the LM limit corresponding to aimed feasible blocking scheme. The process is shown in Fig. 5.

2) The Base Curve has not Passed the LM Limit: If the *Base Curve* reaches its SNB before the LM limit, the starting state point, $\boldsymbol{x}_{base}$, in scenario 1 does not exist anymore as shown in Fig. 6. We take the *Base Curve* SNB, $\boldsymbol{x}_{nose,base}$, to be the starting point instead. Because this is the closest point to the LM limit on the *Base Curve*. Since this point is generally still far away from $\boldsymbol{x}_{aimed}$, there is a larger chance that the NR process encounters divergence. Therefore, we introduce a middle point $\boldsymbol{x}_{aimed,middle}$ on the aimed curve to be the goal of the NR process. The reason is that this point has the largest chance to

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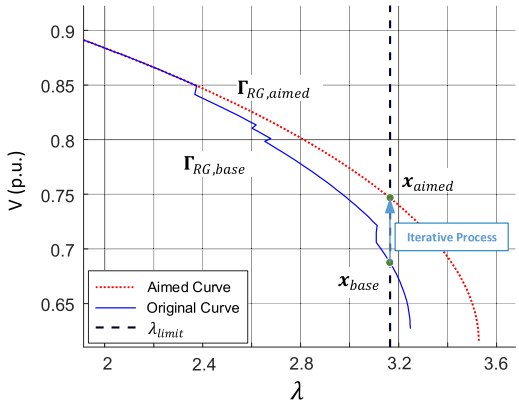


Fig. 5. RG voltage magnitude estimation on feasible curves – Scenario 1.

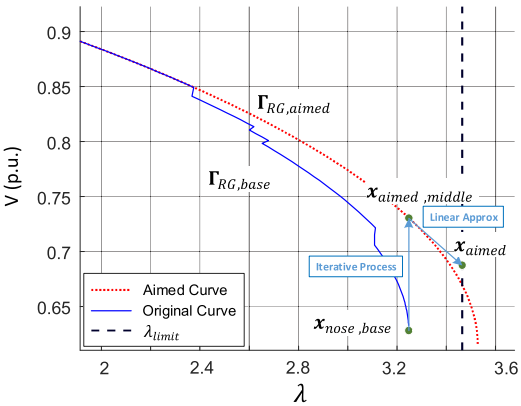


Fig. 6. RG voltage magnitude estimation on feasible curves – Scenario 2.

be both close to the aimed point, $\boldsymbol{x}_{aimed}$. It is also on the aimed curve which could introduce the least estimation error. We use tangent estimation based on the gradient on $\boldsymbol{x}_{aimed,middle}$ to find the estimated $\boldsymbol{x}_{aimed}$.

With the $\boldsymbol{V}$ estimation at $\boldsymbol{x}_{aimed}$, we can estimate the decrease of LVRT settings for the current blocking scheme by,

$$\Delta\, \boldsymbol{c}_{LVRT} = \left[\max\left\{\left(\boldsymbol{c}_{LVRT}^{original}(i) - \boldsymbol{V}_{RG}(i)\right), 0\right\}\right], \quad \forall\, i \in \{1, \ldots, N_{RG}\} \tag{9}$$

where $\boldsymbol{V}_{RG}$ is the vector of RG voltage magnitudes at $\boldsymbol{x}_{aimed}$.

The estimated candidate LVRT adjustment solutions corresponding to each feasible blocking scheme are thus found in the resulting matrix $\boldsymbol{D}$. Detailed algorithm is in Algorithm 3.

D. Optimum Adjustment Solution Selection and Evaluation

With the resulted candidate solution $\Delta \boldsymbol{c}_{LVRT}$ for each feasible blocking scheme, it is easy to rank them w.r.t the cumulative LVRT adjustments. However, these setting adjustments and the LM increment for the feasibility test is find by estimation. So a complete CPFLOW is necessary for each interested blocking scheme to find the exact LVRT adjustments to realize the blocking and find the true LM after the adjustment to justify its feasibility.

Algorithm 3: LVRT Limits Adjustment Estimation.

Input: Feasible blocking schemes set, $\mathbb{B}_{feas}$; Base Curves set, $\mathbb{K}$; RG bus LVRT limits set, $\mathbb{C}$; $\boldsymbol{\lambda}_{limit}$
Output: Estimated LVRT limits adjustment solution matrix, $\boldsymbol{D}$ Loop through feasible blocking schemes:

```
find Base Curve RG final statuses, Γ_{RG,base}
for i^{th} scheme ∈ B_{feas} do
    find corresponding RG connection statuses, Γ_{RG,i}
    if λ*_{base} ≥ λ_{limit} then
        acquire initial state vector, x_{base}, on Base Curves at λ_{limit}
        use Newton-Raphson method to iterate from x_{base} on Γ_{RG,base} curves to x_{aimed} on Γ_{RG,i} curves
    else
        acquire initial state vector, x_{nose,base} on Base Curves at λ*_{base}
        use Newton-Raphson method to iterate from x_{nose,base} on Γ_{RG,base} curves to x_{aimed,middle} on Γ_{RG,i} curves
        find Jacobian matrix, J_i, at x_{aimed,middle}
        use J_i and λ*_{base} to estimate x_{aimed} at λ_{limit}
    end if
    acquire RG bus voltage vector, V_{RG} from x_{aimed}
    calculate LVRT limit adjustments Δc_{LVRT,i}, using (9)
    append Δc_{LVRT,i} to D
end for
return B_{feas}
```

However, at this stage, a large part of the candidate solutions have been ruled out through criticality and feasibility. The efficiency of the whole method is not compromised by conducting CPFLOW on a certain few highest-ranked candidate solutions. As aforementioned, this ranking by minimal total LVRT adjustment is in consideration of bulk power system security. While it is more of an intuitive metric used in this paper, other quantitative methods such as weighted total LVRT adjustment are also applicable. But the basic idea of seeking most system operation security should be preserved. The number of selected candidate solutions, m, is also a user-defined variable. The process is shown in Algorithm 4.

V. Case Study and Discussion

The proposed method is tested on the IEEE 14-bus system, the IEEE 118-bus system, and a synthetic South Carolina 500-bus power system model[35]. We modified each of them to add a high RG penetration. The results are shown in detail below and the effectiveness and efficiency of the proposed method is discussed.

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TABLE I
RG SETTINGS OF IEEE 14-BUS CASE

RG Bus Number	4	5	9	10	11	12	13	14
PG (p.u.)	0.4	0.07	0.2	0.09	0.035	0.05	0.1	0.1
LVRT Const.	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9

TABLE II
BASE CURVE EVENTS OF IEEE 14-BUS CASE

Event Number	Load Scale (λ)	Event Bus Number	Event Type
1	1.8559	14	RG Trip
2	1.9673	9	RG Trip
3	1.9673	10	RG Trip
4	2.5730	4	RG Trip
5	2.5730	2	Gen Bus Q_{max} Limit Hit
6	2.5730	5	RG Trip
7	2.7157	-	Curve Reaching SNB

A. IEEE 14-Bus System

The IEEE 14-bus test system has 5 generation buses and 9 load buses. We added 8 RGs to the load buses with details provided in Table I.

Algorithm 4: Top Candidate Solution Evaluation.

Input: Estimated LVRT limits adjustment solution matrix, $\boldsymbol{D}$; number of selected candidate solutions, $\boldsymbol{m}$
Output: Resulted LVRT limits adjustment solutions, $\boldsymbol{D_{actual}}$

Loop through candidate solutions:
1: sum up estimated LVRT adjustments in each candidate solution and rank
2: find top $\boldsymbol{m}$ solutions in the ranking results with minimum total adjustments and list them in $\boldsymbol{D_{estimate}}$
Loop through top m candidate solutions:
3: **for** $\boldsymbol{i} = 1 : \boldsymbol{m}$ **do**
4: current candidate solution is $\boldsymbol{D_{estimate}(i)}$
5: block RGs with positive LVRT adjustment in $\boldsymbol{D_{estimate}(i)}$ from tripping and run CPFLOW
6: find actual LVRT adjustments for these RGs and save in $\boldsymbol{D_{actual}(i)}$
7: **end for**
8: **return** $\boldsymbol{D_{actual}}$

On the *Base Curve*, 6 RGs trip events and one generation bus reaches Q limit before reaching the SNB as shown in Table II. The LM of the *Base Curve* is 2.7157. Let the additional LM required be 0.3, i.e., $\lambda_{\text{limit}} = 3.0157$.

The Critical RG buses identified using the method proposed in Section IV. A for LVRT adjustments are bus 4, 5, 9, 10, 14 identified with the method proposed in Section IV. A. The *Base Curve* with these events as well as two example (Bus 5 and Bus 11) voltage prediction results are illustrated in Fig. 7. Each example represents a group of critical or non-critical RGs,

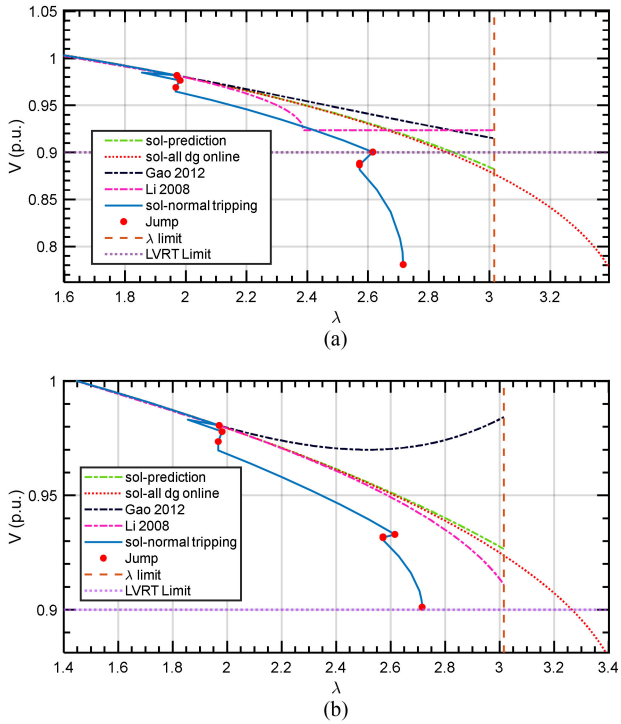


Fig. 7. RG voltage prediction results examples of 14-bus case. (a) RG bus voltage prediction example: Critical RG on bus 5. (b) RG bus voltage prediction example: Non-critical RG on bus 11.

TABLE III
DATA POINTS FOR VOLTAGE PREDICTION OF TWO EXAMPLE RG BUSES IN IEEE 14-BUS CASE

V at Bus 5	V at Bus 11	Load Scale (λ)
1.0584	1.0410	0.0000
1.0145	1.0026	1.3723
1.0046	0.9956	1.5753
0.9975	0.9908	1.7078
0.9938	0.9884	1.7731
0.9900	0.9859	1.8378
0.8820	0.9269	3.0157

respectively. Data points used for LSE method are shown in Table III, where the last row is prediction at λ_{limit}. The estimated parameters for Bus 5 are [0.0062, −0.1629, 1.0061, 0.2044, −0.5956, −0.4966]$^{\mathbf{T}}$, and the parameters for Bus 11 are [−0.0009, −0.1724, 0.9983, 0.2150, −0.5726, −0.4856]$^{\mathbf{T}}$. The accuracy for the voltage prediction method is illustrated by these example plots. The prediction errors are negligible in most cases. Even when the true voltage value is close to the LVRT limit like Fig. 7(a), the algorithm is capable of providing accurate enough prediction and further correct identification results. 5 out of 8 RGs in this case are identified as critical. This single step compresses the searching space from $2^8 = 256$ to $2^5 = 32$.

The best three blocking schemes are selected and evaluated with the full CPFLOW. The estimated LVRT setting adjustments and LM increments and their actual values acquired by CPFLOW are listed in Table IV for comparison. The best scheme is labeled red. From these results, we can see that the estimated values are very close to the actual ones. Even though

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TABLE IV
RG LVRT ADJUSTMENT SOLUTIONS W/ LOWEST TOTAL ADJUSTMENT OF IEEE 14-BUS CASE

Top Solution No.		$\Delta c_{LVRT} = c_{LVRT}^{original} - c_{LVRT}^{new}$					$\sum \Delta c_{LVRT}$	LM Increment
		Bus 4	Bus 5	Bus 9	Bus 10	Bus 14		
1	Estimate	0.0334	0.0310	0.1104	0	0.1283	0.3031	0.3203
	Actual	0.0376	0.0399	0.1385	0	0.1503	0.3663	0.3135
2	Estimate	0.0251	0	0.1011	0.0873	0.1216	0.3351	0.3466
	Actual	0.0272	0	0.1114	0.1049	0.1376	0.3811	0.3726
3	Estimate	0.0199	0.0166	0.0984	0.0850	0.1197	0.3396	0.3664
	Actual	0.0170	0.0165	0.1159	0.1002	0.1335	0.3831	0.3906

for some cases, there do exist differences between the estimated and the actual numerical results, the rank order of the actual adjustment schemes remains consistent with the estimated ones. These differences should not matter much due to the fact that the final output of the whole process is the CPFLOW evaluated actual scheme. As long as the best blocking scheme falls in the range of solutions taken to the evaluation step, it will surely be found through the CPFLOW. In the worst scenario, 3 schemes might not include the best blocking scheme, users can simply increase the number of selected schemes for evaluation. Even the selected cases number reaches 20, it is still not comparable to hundreds or even millions of full CPFLOW computation we used to face in large scale system cases.

The total processing time for the proposed method is 9.4044 seconds under hardware environment of Intel(R) Core(TM) i9-9900T CPU @ 2.10GHz 32 GB RAM. The time consumption for *Base Curve* calculation is 3.3046 seconds, time for critical RG identification is 1.1168 seconds, time for screening is 1.6337 seconds, and time for evaluation is 3.3461 seconds. The screening step is the core of the proposed method and it is efficient thanks to the approximations embedded. The most time consuming part is the update of system Jacobian matrix when the generator reaches Q limit and bus type changes. This is because we realized the computation by generating a MATLAB function based on the symbolic expressions of the Jacobian matrix for reusability. This function needs to be regenerated once the Jacobian matrix is updated.

Using brute-force method to evaluate every possible blocking scheme (256 in total) under the same case, the results exactly match those obtained by the proposed method, which validates its effectiveness. For the brute-force method, the total time is 18.0822 seconds.

B. IEEE 118-Bus System

For this case, 32 RGs are placed, one on each load bus of the IEEE 118-bus test system. The RG bus number, capacity, and original LVRT settings are listed in the Appendix. The capacities of added RGs are set according to the local load demand at the corresponding bus, largely making up for at least 50% of the demand. The original LVRT settings are also set differently for different locations in the system. On the *Base Curve*, there are 9 events before the system reaches the SNB, which are shown in Table V. The LM of the *Base Curve* is 2.5237. The desired LM $\lambda_{\text{limit}} = 2.5837$.

TABLE V
BASE CURVE EVENTS OF IEEE 118-BUS CASE

Event Number	Load Scale (λ)	Event Bus Number	Event Type
1	1.1815	44	RG Trip
2	1.4366	45	RG Trip
3	1.5005	21	RG Trip
4	1.5771	22	RG Trip
5	1.6552	52	RG Trip
6	1.7626	43	RG Trip
7	2.044	51	RG Trip
8	2.3236	8	Gen Bus Q_{max} Limit Hit
9	2.3553	49	Gen Bus Q_{max} Limit Hit
10	2.5237	-	Curve Reaching SNB

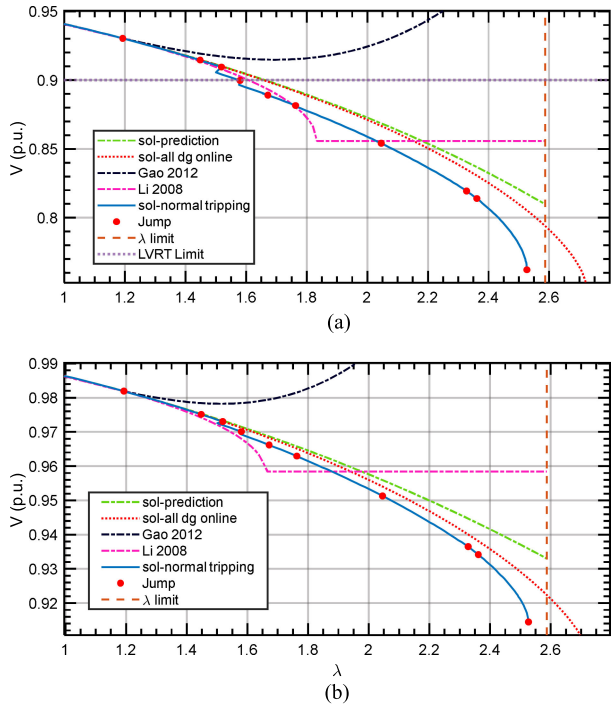


Fig. 8. RG voltage prediction results examples of 118-bus case. (a) RG bus voltage prediction example: Critical RG on bus 22. (b) RG bus voltage prediction example: Non-critical RG on bus 23.

Critical RG buses identified are 21, 22, 43, 44, 45, 51, 52 according to the base P-V curve. The *Base Curve* and two example voltage prediction results are illustrated in Fig. 8, where the LVRT constant is set to be 0.8 for the non-critical bus 23, therefore not appearing in Fig. 8(b). The efficacy of the prediction method is not effectively affected by the scale of the

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TABLE VI
RG LVRT ADJUSTMENT SOLUTIONS W/ LOWEST TOTAL ADJUSTMENT OF IEEE 118-BUS CASE

Top Solution No.		$\Delta c_{LVRT} = c_{LVRT}^{original} - c_{LVRT}^{new}$							$\sum \Delta c_{LVRT}$	LM Increment
		Bus 21	Bus 22	Bus 43	Bus 44	Bus 45	Bus 51	Bus 52		
1	Estimate	0.1204	0.1177	0.0836	0	0.1524	0.0485	0	0.5226	0.0605
	Actual	0.1205	0.1177	0.0846	0	0.1544	0.0494	0	0.5266	0.0664
2	Estimate	0.1204	0.1176	0.0836	0	0.1522	0	0.0679	0.5417	0.0606
	Actual	0.1204	0.1177	0.0846	0	0.1542	0	0.0687	0.5456	0.0666
3	Estimate	0.1198	0.1170	0.0834	0	0.1513	0.0437	0.0646	0.5798	0.0616
	Actual	0.1199	0.1171	0.0843	0	0.1533	0.0446	0.0653	0.5845	0.0681
4	Estimate	0.1218	0	0.0831	0.1913	0.1475	0.0484	0	0.5921	0.0600
	Actual	0.1220	0	0.0842	0.1935	0.1495	0.0493	0	0.5985	0.0647
5	Estimate	0.1218	0	0.0831	0.1912	0.1474	0	0.0679	0.6114	0.0601
	Actual	0.1219	0	0.0842	0.1934	0.1494	0	0.0687	0.6176	0.0649

system. In this case, with 7 out of RGs identified as critical, the searching space shrinks from $2^{32} = 4294967296$ to $2^7 = 128$. This is a substantial advantage when considering a large RG fleet.

The best 5 blocking schemes are selected and evaluated against the full CPFLOW. The estimated and actual LVRT setting adjustments and LM increments are listed in Table VI. The best scheme is labeled red. These results are also validated using the brute-force method. It is worth noting that the complete search space in this case has a dimension of $2^{32} \times 32$ (1024.0GB) which is far beyond the allowed maximum matrix size and memory capacity. Furthermore, the processing time makes it impossible to apply in practice. Therefore, the brute-force method was only tested on a subset of the search space, which is spanned by the identified critical RGs. The total processing time for the proposed method is 125.2008 seconds. The core screening step takes 16.2110 seconds. In comparison, the brute-force method requires 5233.7128 seconds in total.

TABLE VII
BASE CURVE EVENTS OF 500-BUS CASE

Event Number	Load Scale (λ)	Event Bus Number	Event Type
1	1.2187	28	RG Trip
2	1.3641	98	RG Trip
3	1.401	9	Gen Bus Q_{max} Limit Hit
4	1.405	69	RG Trip
5	1.4787	144	Gen Bus Q_{max} Limit Hit
6	1.5488	142	RG Trip
7	1.5522	128	Gen Bus Q_{max} Limit Hit
8	1.5522	88	RG Trip
9	1.5701	122	RG Trip
10	1.5701	30	RG Trip
11	1.5701	105	RG Trip
12	1.5701	32	RG Trip
13	1.5701	51	RG Trip
14	1.5701	52	RG Trip
15	1.5701	54	RG Trip
16	1.5701	75	RG Trip
17	1.5701	103	RG Trip
18	1.5768	-	Curve Reaching SNB

C. Synthetic 500-Bus System

In this case, 1 RG is added to each of the 64 load buses. The RG capacities are set up in the same manner as in the IEEE 118-bus system. The original LVRT constants are all set to be 0.75 in order to mimic a realistic scenario where a utility uses a default value throughout their system. The *Base Curve* comprises of 17 events before reaching the SNB as shown in Table VII. The LM of the *Base Curve* is 1.5768. The desired load margin $\lambda_{\text{limit}} = 1.69$.

21 RG buses are identified as critical. These comprise of bus 6, 36, 59, 86, 115, 116, 130 along with the ones tripping on the base curve. The best 5 blocking schemes are selected for further evaluation. For these, the estimated and actual LVRT setting adjustments and LM increments are listed in Table VIII. Due to the comparatively large number of critical RGs and limited space, several critical RGs, which did not require LVRT adjustments are omitted from the result table.

We can see from the table that the differences of LVRT adjustments on the same RG among the top three schemes are generally small, so is the LM increment. This is mostly due to the large scale of the system and the number of RGs. The capacity of a single RG is comparatively small. With one or two more RGs blocked, they are not effectively affecting other RG's voltage profile. Single RG's impact on the LM is also generally not phenomenal. This is reflected with nearby P-V curves and thus close LMs. Besides, the increase of system scale is not negatively impacting the estimation accuracy. The estimation errors in the 500-bus system are even smaller than in the 14-bus system. The reason could be that in smaller-scale systems, load increase can cause a more significant voltage decrease when close to LM and thus more curvature in the P-V curve, which makes the linear estimation less effective at $\lambda = \lambda_{\text{limit}}$.

The core screening step takes 411.5768 seconds in this case. Despite the long processing time, it is worth noting that the proposed method makes this problem solvable. It is not realistic or feasible to find an adjustment solution for this case using brute force method which involves 2^{64} times full CPFLOW. There is also much improvement space given the major part of the time consumption is caused by the aforementioned Jacobian update process when a generator reaches Q limit. It averagely needs around 583 seconds to regenerate the MATLAB function each time the Jacobian matrix is updated. This can be avoided in future industrial realizations. In addition, the approach can easily be parallelized in critical RG identification and evaluation steps. More advanced computation resources which can be

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TABLE VIII
RG LVRT ADJUSTMENT SOLUTIONS W/ LOWEST TOTAL ADJUSTMENT OF 500-BUS CASE

Top Solution No.		$\Delta c_{LVRT} = c_{LVRT}^{original} - c_{LVRT}^{new}$											LM Increment
		Bus 28	Bus 30	Bus 32	Bus 51	Bus 52	Bus 54	Bus 69	Bus 75	Bus 98	Bus 103	Bus 105	Bus 122	
1	Estimate	0.1475	0.0581	0.0188	0	0	0.0324	0.0464	0	0.1287	0	0.0229	0.0640	0.1142
	Actual	0.1476	0.0582	0.0188	0	0	0.0325	0.0465	0	0.1287	0	0.0229	0.0640	0.1181
2	Estimate	0.1475	0.0578	0.0182	0	0	0.0322	0.0464	0.0102	0.1287	0	0.0223	0.0637	0.1144
	Actual	0.1475	0.0580	0.0184	0	0	0.0323	0.0464	0.0104	0.1288	0	0.0226	0.0638	0.1187
3	Estimate	0.1476	0.0580	0.0185	0	0	0.0323	0.0466	0.0105	0.1287	0.0814	0.0226	0.0639	0.1147
	Actual	0.1475	0.0580	0.0186	0	0	0.0323	0.0464	0.0106	0.1287	0.0816	0.0228	0.0639	0.1192
4	Estimate	0.1698	0.0627	0.0297	0	0.0868	0.0369	0.0620	0.0207	0.2676	0.0814	0.0352	0.0688	0.1148
	Actual	0.1700	0.0628	0.0297	0	0.0868	0.0370	0.0622	0.0207	0.2676	0.0815	0.0352	0.0689	0.1196
5	Estimate	0.1697	0.0626	0.0297	0.0855	0.0868	0.0368	0.0619	0.0206	0.2677	0.0814	0.0351	0.0687	0.1150
	Actual	0.1697	0.0628	0.0299	0.0854	0.0867	0.0369	0.0619	0.0209	0.2676	0.0814	0.0354	0.0688	0.1201

easily accessed by utilities can significantly improve the time consumption.

Given the events that happened in the base P-V curves of all three cases, we can find that with the increase of RGs installed in systems of various scales, original standard LVRT setting violations are no longer rare when system loading increment. Events leading to system collapse like aforementioned blocking have also occurred within experiments by the authors [31]. These facts further certify the necessity of the proposed method.

The results of three study cases demonstrate the effectiveness of each step of the proposed method. With the base P-V curve, critical RGs are identified by predicting their voltage magnitudes at designated λ_{limit}. The LM increment estimation method can provide reliable results in each case to facilitate blocking schemes feasibility check. The LVRT adjustment estimation algorithm is also capable of finding corresponding adjustment quantities. Given these major steps validated, the proposed method is fully capable of finding the LVRT settings adjustment scheme adaptively for systems of various scales, which can help increase system LM and thus improve voltage security with the least risk.

VI. CONCLUSION

A scalable multi-stage method is proposed in this paper to adapt RGs' LVRT settings as a preventive control measure to enhance voltage security. The overall process comprises of a CPFLOW with loss of RGs accounted for, identification of critical RGs using a proposed PV curve approximation method, screening feasible candidates using LM sensitivity, estimation of minimum required LVRT adjustments, and an optimal RG blocking schemes selection and evaluation. The main contribution lies in the formulation of the LVRT adjustment problem. Other contributions include proposing a novel voltage prediction method, an estimation-based scalable scheme for feasible RG operation limits adjustment solution screening, and application of the proposed method in large scale renewable-rich system study cases. The overall goal is to help the system attain a desired LM by preventing nuisance RG trips due to LVRT violation. By avoiding repetitive CPFLOW calculation, the approach is scalable and can be used both online to adapt LVRT settings as well as offline to identify areas requiring dynamic VAR support to minimize harmful loss of generation scenarios. Three study cases with various scales are used to demonstrate the effectiveness and efficiency of the method.

APPENDIX

The RG integration in the IEEE 118-bus case is as follows:

TABLE IX
RG SETTINGS OF IEEE 118-BUS CASE

RG Bus Number	2	3	7	11	13	14	16	17
PG (p.u.)	0.2	0.39	0.19	0.7	0.34	0.14	0.25	0.11
LVRT Const.	0.9	0.9	0.9	0.9	0.9	0.8	0.9	0.8
RG Bus Number	20	21	22	23	28	29	33	35
PG (p.u.)	0.18	0.14	0.1	0.07	0.085	0.24	0.23	0.165
LVRT Const.	0.8	0.9	0.9	0.8	0.9	0.8	0.8	0.9
RG Bus Number	39	41	43	44	45	47	48	50
PG (p.u.)	0.27	0.185	0.18	0.08	0.265	0.17	0.1	0.085
LVRT Const.	0.8	0.9	0.9	0.9	0.9	0.8	0.9	0.8
RG Bus Number	51	52	53	57	58	79	82	84
PG (p.u.)	0.085	0.09	0.115	0.06	0.06	0.195	0.27	0.055
LVRT Const.	0.9	0.9	0.8	0.8	0.8	0.8	0.8	0.8

REFERENCES

[1] IEA, "Global energy review 2020 – analysis," Accessed: Jul./May, 2020. [Online]. Available: https://www.iea.org/reports/global-energy-review-2020/renewables

[2] *IEEE Standard for Interconnection and Interoperability of Distributed Energy Resources With Associated Electric Power Systems Interfaces*, IEEE Standard 1547-2018 Revised IEEE Standard 1547-2003, pp. 1–138, Apr. 2018.

[3] J. Kim, C.-H. Kim, Y. Oh, G.-J. Cho, and J.-S. Song, "An islanding detection method for multi-RES systems using the graph search method," *IEEE Trans. Sustain. Energy*, vol. 11, no. 4, pp. 2722–2731, Oct. 2020.

[4] K. Jia, H. Wei, T. Bi, D. W. P. Thomas, and M. Sumner, "An islanding detection method for multi-DG systems based on high-frequency impedance estimation," *IEEE Trans. Sustain. Energy*, vol. 8, no. 1, pp. 74–83, Jan. 2017.

[5] *UKERC*, "What happened to our electricity system on friday august 9th 2019?," Accessed: Dec. 30, 2020. [Online]. Available: https://ukerc.ac.uk/news/what-happened-electricity-system-fri-aug-9-2019/

[6] C. Mishra, A. Pal, J. S. Thorp, and V. A. Centeno, "Transient stability assessment of prone-to-trip renewable generation rich power systems using Lyapunov's direct method," *IEEE Trans. Sustain. Energy*, vol. 10, no. 3, pp. 1523–1533, Jul. 2019.

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[7] C. Wang, C. Mishra, and V. A. Centeno, "Low voltage ride through (LVRT) constrained transient stability assessment using Lyapunov functions family method," in *Proc. IEEE Power Energy Soc. Gen. Meeting*, 2020, pp. 1–5, doi: 10.1109/PESGM41954.2020.9281525.

[8] V. Ajjarapu and C. Christy, "The continuation power flow: A tool for steady state voltage stability analysis," in *Proc. Conf. Papers Power Ind. Comput. Appl. Conf.*, 1991, pp. 304–311, doi: 10.1109/PICA.1991.160593.

[9] L. Wang and H. Chiang, "Toward online bus-bar splitting for increasing load margins to static stability limit," *IEEE Trans. Power Syst.*, vol. 32, no. 5, pp. 3715–3725, Sep. 2017.

[10] A. Trias, "The holomorphic embedding load flow method," in *Proc. IEEE Power Energy Soc. Gen. Meeting*, 2012, pp. 1–8, doi: 10.1109/PESGM.2012.6344759.

[11] Y. Xu, L. Mili, M. Korkali, K. Karra, Z. Zheng, and X. Chen, "A data-driven nonparametric approach for probabilistic load-margin assessment considering wind power penetration," *IEEE Trans. Power Syst.*, vol. 35, no. 6, pp. 4756–4768, Nov. 2020.

[12] M. E. C. Bento and R. A. Ramos, "An approach for monitoring and updating the load margin of power systems in dynamic security assessment," *Electr. Power Syst. Res.*, vol. 198, Sep. 2021, Art. no. 107365, doi: 10.1016/j.epsr.2021.107365.

[13] A. Chandra and A. K. Pradhan, "Online voltage stability and load margin assessment using wide area measurements," *Int. J. Electr. Power Energy Syst.*, vol. 108, pp. 392–401, Jun. 2019, doi: 10.1016/j.ijepes.2019.01.021.

[14] F. Dong, B. H. Chowdhury, M. L. Crow, and L. Acar, "Improving voltage stability by reactive power reserve management," *IEEE Trans. Power Syst.*, vol. 20, no. 1, pp. 338–345, Feb. 2005.

[15] A. Rabiee, S. M. Mohseni-Bonab, M. Parniani, and I. Kamwa, "Optimal cost of voltage security control using voltage dependent load models in presence of demand response," *IEEE Trans. Smart Grid*, vol. 10, no. 3, pp. 2383–2395, May 2019.

[16] Y.-T. Hsiao, C.-C. Liu, H.-D. Chiang, and Y.-L. Chen, "A new approach for optimal VAR sources planning in large scale electric power systems," *IEEE Trans. Power Syst.*, vol. 8, no. 3, pp. 988–996, Aug. 1993.

[17] T. Niu, Q. Guo, H. Sun, B. Wang, and B. Zhang, "Voltage security regions considering wind power curtailment to prevent cascading trip faults in wind power integration areas," *IET Renewable Power Gener.*, vol. 11, no. 1, pp. 54–62, Aug. 2016, doi: 10.1049/iet-rpg.2016.0151.

[18] H. M. Hasanien, "An adaptive control strategy for low voltage ride through capability enhancement of grid-connected photovoltaic power plants," *IEEE Trans. Power Syst.*, vol. 31, no. 4, pp. 3230–3237, Jul. 2016.

[19] Y. M. Alsmadi *et al.*, "Detailed investigation and performance improvement of the dynamic behavior of grid-connected DFIG-based wind turbines under LVRT conditions," *IEEE Trans. Ind. Appl.*, vol. 54, no. 5, pp. 4795–4812, Sep. 2018.

[20] J. Chen, L. Jiang, W. Yao, and Q. H. Wu, "Perturbation estimation based nonlinear adaptive control of a full-rated converter wind turbine for fault ride-through capability enhancement," *IEEE Trans. Power Syst.*, vol. 29, no. 6, pp. 2733–2743, Nov. 2014.

[21] D. Xie, Z. Xu, L. Yang, J. Østergaard, Y. Xue, and K. P. Wong, "A comprehensive LVRT control strategy for DFIG wind turbines with enhanced reactive power support," *IEEE Trans. Power Syst.*, vol. 28, no. 3, pp. 3302–3310, Aug. 2013.

[22] J. Ouyang, T. Tang, J. Yao, and M. Li, "Active voltage control for DFIG-Based wind farm integrated power system by coordinating active and reactive powers under wind speed variations," *IEEE Trans. Energy Convers.*, vol. 34, no. 3, pp. 1504–1511, Sep. 2019.

[23] R. Liu, J. Yao, X. Wang, P. Sun, J. Pei, and J. Hu, "Dynamic stability analysis and improved LVRT schemes of DFIG-based wind turbines during a symmetrical fault in a weak grid," *IEEE Trans. Power Electron.*, vol. 35, no. 1, pp. 303–318, Jan. 2020.

[24] M. Easley, S. Jain, M. B. Shadmand, and H. A. Abu-Rub, "Autonomous model predictive controlled smart inverter with proactive grid fault ride-through capability," *IEEE Trans. Energy Convers.*, vol. 35, no. 4, pp. 1825–1836, Dec. 2020.

[25] W. Guo, L. Xiao, and S. Dai, "Enhancing low-voltage ride-through capability and smoothing output power of DFIG with a superconducting fault-current limiter–magnetic energy storage system," *IEEE Trans. Energy Convers.*, vol. 27, no. 2, pp. 277–295, Jun. 2012.

[26] P. Mukherjee and V. V. Rao, "Superconducting magnetic energy storage for stabilizing grid integrated with wind power generation systems," *J. Mod. Power Syst. Clean Energy*, vol. 7, no. 2, pp. 400–411, Mar. 2019, doi: 10.1007/s40565-018-0460-y.

[27] M. Firouzi and G. B. Gharehpetian, "LVRT performance enhancement of DFIG-based wind farms by capacitive bridge-type fault current limiter," *IEEE Trans. Sustain. Energy*, vol. 9, no. 3, pp. 1118–1125, Jul. 2018.

[28] Z.-X. Zheng, C.-J. Huang, R.-H. Yang, X.-Y. Xiao, and C.-S. Li, "A low voltage ride through scheme for DFIG-based wind farm with SFCL and RSC control," *IEEE Trans. Appl. Supercond.*, vol. 29, no. 2, pp. 1–5, Mar. 2019.

[29] W. A. Qureshi and N. C. Nair, "Systematic development of low voltage ride-through (LVRT) envelope for grids," in *Proc. IEEE Region Conf.*, 2010, pp. 574–579, doi: 10.1109/TENCON.2010.5686749.

[30] C. Rahmann, H.-J. Haubrich, A. Moser, R. Palma-Behnke, L. Vargas, and M. B. C. Salles, "Justified fault-ride-through requirements for wind turbines in power systems," *IEEE Trans. Power Syst.*, vol. 26, no. 3, pp. 1555–1563, Aug. 2011.

[31] C. Wang, C. Mishra, R. S. Biswas, A. Pal, and V. A. Centeno, "Adaptive LVRT settings adjustment for enhancing voltage security of renewable-rich electric grids," in *Proc. IEEE Power Energy Soc. Gen. Meeting*, 2020, pp. 1–5, doi: 10.1109/PESGM41954.2020.9282044.

[32] F. Gao, J. S. Thorp, A. Pal, and S. Gao, "Dynamic state prediction based on auto-regressive (AR) model using PMU data," in *Proc. IEEE Power Energy Conf. Illinois*, IL, USA, 2012, pp. 1–5, doi: 10.1109/PECI.2012.6184586.

[33] S.-H. Li and H.-D. Chiang, "Look-ahead Q-constrained load margin for large-scale power systems," in *Proc. IEEE Power Energy Soc. Gen. Meeting - Convers. Del. Elect. Energy 21st Century*, 2008, pp. 1–6, doi: 10.1109/PES.2008.4596718.

[34] S. Greene, I. Dobson, and F. L. Alvarado, "Sensitivity of the loading margin to voltage collapse with respect to arbitrary parameters," *IEEE Trans. Power Syst.*, vol. 12, no. 1, pp. 262–272, Feb. 1997.

[35] A. B. Birchfield, T. Xu, K. M. Gegner, K. S. Shetye, and T. J. Overbye, "Grid structural characteristics as validation criteria for synthetic networks," *IEEE Trans. Power Syst.*, vol. 32, no. 4, pp. 3258–3265, Jul. 2017.



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